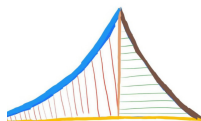


Math for EE

1



G. V. V. Sharma

ABOUT THIS BOOK

This book introduces matrices through high school coordinate geometry. This approach makes it easier for beginners to learn Python for scientific computing. All problems in the book are from NCERT mathematics textbooks from Class 9-12. Exercises are from CBSE board exam papers.

The content is sufficient for industry jobs and covers nearly all matrix prerequisites for machine learning. There is no copyright, so readers are free to print and share.

License: <https://creativecommons.org/licenses/by-sa/3.0/>
and

<https://www.gnu.org/licenses/fdl-1.3.en.html>

First manual appeared in January 2018

First edition published on July 10, 2024

In this edition, some incorrect solutions were removed. All figures redrawn. More problems added.

CONTENTS

1	GATE DA 2025	4
1.1	Problems	5

1. GATE DA 2025

A. Problems

- 1.1.1 A 4×4 digital image has pixel intensities (U) as shown in the figure. The number of pixels with $U \leq 4$ is: (DA 2025)

0	1	0	2
4	7	3	3
5	5	4	4
6	7	3	2

Fig. 1.1.1.1: Pixel Grid

- a) 3 b) 8 c) 11 d) 9
- 1.1.2 A rectangle has a length L and a width W , where $L > W$. If the width, W , is increased by 10%, which one of the following statements is correct for all values of L and W ? (DA 2025)
- a) Perimeter increases by 10%.
b) Length of the diagonals increases by 10%.
c) Area increases by 10%.
d) The rectangle becomes a square.
- 1.1.3 Weight of a person can be expressed as a function of their age. The function usually varies from person to person. Suppose this function is identical for two brothers, and it monotonically increases till the age of 50 years and then it monotonically decreases. Let a_1 and a_2 (in years) denote the ages of the brothers and $a_1 < a_2$. Which one of the following statements is correct about their age on the day when they attain the same weight? (DA 2025)
- a) $a_1 < a_2 < 50$ c) $50 < a_1 < a_2$
b) $a_1 < 50 < a_2$ d) Either $a_1 = 50$ or $a_2 = 50$
- 1.1.4 If a real variable x satisfies $3^{x^2} = 27 \times 9^x$, then the value of $\frac{2^{x^2}}{(2^x)^2}$ is: (DA 2025)
- a) 2^{-1} b) 2^0 c) 2^3 d) 2^{15}
- 1.1.5 Suppose X and Y are random variables. The conditional expectation of X given Y is denoted by $E[X | Y]$. Then $E[E[X | Y]]$ equals (DA 2025)
- a) $E[X | Y]$ b) $\frac{E[X]}{E[Y]}$ c) $E[X]$ d) $E[Y]$
- 1.1.6 The number of additions and multiplications involved in performing Gaussian elimination on any $n \times n$ upper triangular matrix is of the order (DA 2025)

- a) $O(n)$ b) $O(n^2)$ c) $O(n^3)$ d) $O(n^4)$

1.1.7 The sum of the elements in each row of $\mathbf{A} \in \mathbb{R}^{n \times n}$ is 1. If $\mathbf{B} = \mathbf{A}^3 - 2\mathbf{A}^2 + \mathbf{A}$, which one of the following statements is correct (for $\mathbf{x} \in \mathbb{R}^n$)? (DA 2025)

- a) The equation $\mathbf{B}\mathbf{x} = \mathbf{0}$ has no solution
 b) The equation $\mathbf{B}\mathbf{x} = \mathbf{0}$ has exactly two solutions
 c) The equation $\mathbf{B}\mathbf{x} = \mathbf{0}$ has infinitely many solutions
 d) The equation $\mathbf{B}\mathbf{x} = \mathbf{0}$ has a unique solution

1.1.8 Let $f(x) = \frac{e^x - e^{-x}}{2}$, $x \in \mathbb{R}$. Let $f^{(k)}(a)$ denote the k^{th} derivative of f evaluated at a . What is the value of $f^{(10)}(0)$? (Note: ! denotes factorial) (DA 2025)

- a) 0 b) 1 c) $\frac{1}{10!}$ d) $\frac{2}{10!}$

1.1.9 Consider the following three relations:

Car (model, year, serial, color)

Make (maker, model)

Own (owner, serial)

A tuple in Car represents a specific car of a given model, made in a given year, with a serial number and a color. A tuple in Make specifies that a maker company makes cars of a certain model. A tuple in Own specifies that an owner owns the car with a given serial number. Keys are underlined; (owner, serial) together form key for Own. ($\bowtie$ denotes natural join)

$$\pi_{\text{owner}}(\text{Own} \bowtie (\sigma_{\text{color}='red'}(\text{Car} \bowtie (\sigma_{\text{maker}='ABC'}\text{Make})))) \quad (1.1.9.1)$$

Which one of the following options describes what the above expression computes? (DA 2025)

- a) All owners of a red car, a car made by ABC, or a red car made by ABC
 b) All owners of more than one car, where at least one car is red and made by ABC
 c) All owners of a red car made by ABC
 d) All red cars made by ABC

1.1.10 Consider a hash table of size 10 with indices $\{0, 1, \dots, 9\}$, with the hash function

$$h(x) = 3x \pmod{10} \quad (1.1.10.1)$$

where linear probing is used to handle collisions. The hash table is initially empty and then the following sequence of keys is inserted into the hash table: 1, 4, 5, 6, 14, 15. The indices where the keys 14 and 15 are stored are, respectively (DA 2025)

- a) 2 and 5 b) 2 and 6 c) 4 and 5 d) 4 and 6

1.1.11 Let X be a continuous random variable whose cumulative distribution function (CDF)

$F_X(x)$, for some t , is given as follows:

$$F_X(x) = \begin{cases} 0 & x \leq t \\ \frac{x-t}{4-t} & t \leq x \leq 4 \\ 1 & x \geq 4 \end{cases} \quad (1.1.11.1)$$

If the median of X is 3, then what is the value of t ? (DA 2025)

- a) 2 b) 1 c) -1 d) 0

1.1.12 Let $X = aZ + b$, where Z is a standard normal random variable, and a, b are two unknown constants. It is given that

$$E[X] = 1, \quad E[(X - E[X])Z] = -2, \quad E[(X - E[X])^2] = 4, \quad (1.1.12.1)$$

where $E[X]$ denotes the expectation of random variable X . The values of a, b are: (DA 2025)

- a) $a = -2, b = 1$ c) $a = -2, b = -1$
b) $a = 2, b = -1$ d) $a = 1, b = 1$

1.1.13 It is given that $P(X \geq 2) = 0.25$ for an exponentially distributed random variable X with $E[X] = \frac{1}{\lambda}$, where $E[X]$ denotes the expectation of X . What is the value of λ ? ($\ln$ denotes natural logarithm) (DA 2025)

- a) $\ln 2$ b) $\ln 4$ c) $\ln 3$ d) $\ln 0.25$

1.1.14 Consider the following Python declarations of two lists.

```
A = [1, 2, 3]
B = [4, 5, 6]
```

Which one of the following statements results in $A = [1, 2, 3, 4, 5, 6]$? (DA 2025)

- a) `A.extend(B)` b) `A.append(B)` c) `A.update(B)` d) `A.insert(B)`

1.1.15 Consider two functions $f : \mathbb{R} \rightarrow \mathbb{R}$ and $g : \mathbb{R} \rightarrow (1, \infty)$. Both functions are differentiable at a point c . Which of the following functions is/are ALWAYS differentiable at c ? The symbol $\cdot$ denotes product and the symbol $\circ$ denotes composition of functions. (DA 2025)

- a) $f \pm g$ b) $f \cdot g$ c) $\frac{f}{g}$ d) $f \circ g + g \circ f$

1.1.16 Which of the following statements is/are correct? (DA 2025)

- a) $\mathbb{R}^n$ has a unique set of orthonormal basis vectors
b) $\mathbb{R}^n$ does not have a unique set of orthonormal basis vectors

c) Linearly independent vectors in $\mathbb{R}^n$ are orthonormal

d) Orthonormal vectors in $\mathbb{R}^n$ are linearly independent

1.1.17 For which of the following inputs does binary search take time $O(\log n)$ in the worst case? (DA 2025)

a) An array of n integers in any order

b) A linked list of n integers in any order

c) An array of n integers in increasing order

d) A linked list of n integers in increasing order

1.1.18 Let $\mathbf{A} = \mathbf{I}_n + \mathbf{x}\mathbf{x}^\top$ where $\mathbf{I}_n$ is the $n \times n$ identity matrix and $\mathbf{x} \in \mathbb{R}^n$, $\mathbf{x}^\top \mathbf{x} = 1$. Which of the following options is/are correct? (DA 2025)

a) Rank of $\mathbf{A}$ is n

b) $\mathbf{A}$ is invertible

c) 0 is an eigenvalue of $\mathbf{A}$

d) $\mathbf{A}^{-1}$ has a negative eigenvalue

1.1.19 Suppose that insertion sort is applied to the array $[1, 3, 5, 7, 9, 11, x, 15, 13]$ and it takes exactly two swaps to sort the array. Select all possible values of x . (DA 2025)

a) 10

b) 12

c) 14

d) 16

1.1.20 There are three boxes containing white balls and black balls.

- Box-1 contains 2 black and 1 white balls.
- Box-2 contains 1 black and 2 white balls.
- Box-3 contains 3 black and 3 white balls.

In a random experiment, one of these boxes is selected, where the probability of choosing Box-1 is $\frac{1}{2}$, Box-2 is $\frac{1}{6}$, and Box-3 is $\frac{1}{3}$. A ball is drawn at random from the selected box. Given that the ball drawn is white, the probability that it is drawn from Box-2 is _____.

(DA 2025)

1.1.21 $\lim_{t \rightarrow +\infty} \sqrt{t^2 + t} - t = \underline{\hspace{2cm}}$.

(DA 2025)

1.1.22 On a relation named Loan of a bank:

loan_number	branch_name	amount
L11	Banjara Hills	90000
L14	Kondapur	50000
L15	SR Nagar	40000
L22	SR Nagar	25000
L23	Balanagar	80000
L25	Kondapur	70000
L19	SR Nagar	65000

Fig. 1.1.22.1

the following SQL query is executed.

```
SELECT L1.loan_number
```

```
FROM Loan L1
WHERE L1.amount > (SELECT MAX (L2.amount)
FROM Loan L2
WHERE L2.branch_name = 'SR Nagar');
```

The number of rows returned by the query is _____. (DA 2025)

1.1.23 Given data $\{(-1, 1), (2, 5), (3, 5)\}$ of the form (x, y) , we fit a model $y = wx$ using linear least-squares regression. The optimal value of w is _____. (DA 2025)

1.1.24

1.1.25 The naive Bayes classifier is used to solve a two-class classification problem with class-labels y_1, y_2 . Suppose the prior probabilities are $P(y_1) = \frac{1}{3}$ and $P(y_2) = \frac{2}{3}$. Assuming a discrete feature space with

$$P(\mathbf{x} | y_1) = \frac{3}{4} \quad \text{and} \quad P(\mathbf{x} | y_2) = \frac{1}{4} \quad (1.1.25.1)$$

for a specific feature vector $\mathbf{x}$. The probability of misclassifying $\mathbf{x}$ is _____. (DA 2025)

1.1.26 Let $Y = Z^2$, $Z = \frac{X - \mu}{\sigma}$, where X is a normal random variable with mean μ and variance σ^2 . The variance of Y is (DA 2025)

- a) 1 b) 2 c) 3 d) 4

1.1.27 Let $\mathbf{A} \in \mathbb{R}^{n \times n}$ be such that $\mathbf{A}^3 = \mathbf{A}$. Which one of the following statements is ALWAYS correct? (DA 2025)

- a) $\mathbf{A}$ is invertible
b) Determinant of $\mathbf{A}$ is 0
c) The sum of the diagonal elements of $\mathbf{A}$ is 1
d) $\mathbf{A}$ and $\mathbf{A}^2$ have the same rank

1.1.28 Let $\{\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_n\}$ be a set of linearly independent vectors in $\mathbb{R}^n$. Let the (i, j) -th element of matrix $\mathbf{A} \in \mathbb{R}^{n \times n}$ be given by $A_{ij} = \mathbf{x}_i^\top \mathbf{x}_j$, $1 \leq i, j \leq n$. Which one of the following statements is correct? (DA 2025)

- a) $\mathbf{A}$ is invertible
b) 0 is a singular value of $\mathbf{A}$
c) Determinant of $\mathbf{A}$ is 0
d) $\mathbf{z}^\top \mathbf{A} \mathbf{z} = 0$ for some non-zero $\mathbf{z} \in \mathbb{R}^n$

1.1.29 Consider the cumulative distribution function (CDF) of a random variable X :

$$F_X(x) = \begin{cases} 0 & x \leq -1 \\ \frac{1}{4}(x+1)^2 & -1 \leq x \leq 1 \\ 1 & x \geq 1 \end{cases} \quad (1.1.29.1)$$

The value of $P(X^2 \leq 0.25)$ is (DA 2025)

- a) 0.625 b) 0.25 c) 0.5 d) 0.5625

1.1.30 A random variable X is said to be distributed as Bernoulli(θ), denoted by $X \sim \text{Bernoulli}(\theta)$, if $P(X = 1) = \theta$, $P(X = 0) = 1 - \theta$ for $0 < \theta < 1$. Let $Y = \sum_{i=1}^{300} X_i$, where $X_i \sim \text{Bernoulli}(\theta)$, $i = 1, 2, \dots, 300$ be independent and identically distributed random variables with $\theta = 0.25$. The value of $P(60 \leq Y \leq 90)$ after approximation through Central Limit Theorem, is given by (Recall that $\phi(x) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^x e^{-\frac{t^2}{2}} dt$) (DA 2025)

- a) $\phi(2) - \phi(-2)$ c) $\phi(3) - \phi(-3)$
 b) $\phi(1) - \phi(-1)$ d) $\phi(90) - \phi(60)$

1.1.31 For $x \in \mathbb{R}$, the floor function is denoted by $f(x) = \lfloor x \rfloor$ and defined as follows

$$\lfloor x \rfloor = k, \quad k \leq x < k + 1 \quad (1.1.31.1)$$

where k is an integer. Let $Y = \lfloor X \rfloor$ where X is an exponentially distributed random variable with mean $\frac{1}{\ln 10}$, where $\ln$ denotes natural logarithm. For any positive integer l , one can write the probability of the event $Y = l$ as follows

$$P(Y = l) = q^l (1 - q) \quad (1.1.31.2)$$

The value of q is

(DA 2025)

- a) 0.1 b) 0.01 c) 0.5 d) 0.434

1.1.32 A random experiment consists of throwing 100 fair dice, each die having six faces numbered 1 to 6. An event A represents the set of all outcomes where at least one of the dice shows a 1. Then, $P(A) =$ (DA 2025)

- a) 0 b) 1 c) $\left(\frac{5}{6}\right)^{100}$ d) $1 - \left(\frac{5}{6}\right)^{100}$

1.1.33 Consider the following Python code snippet.

```
A = {"this", "that"}
B = {"that", "other"}
C = {"other", "this"}
while "other" in C:
    if "this" in A:
        A, B, C = A - B, B - C, C - A
    if "that" in B:
        A, B, C = C | A, A | B, B | C
```

When the above program is executed, at the end, which of the following sets contains "this"? (DA 2025)

- a) Only A b) Only B c) Only C d) A, C

1.1.34 Consider the function

$$f(x) = \frac{x^3}{3} + \frac{7}{2}x^2 + 10x + \frac{133}{2}, \quad x \in [-8, 0] \quad (1.1.34.1)$$

Which of the following statements is/are correct? (DA 2025)

- a) The maximum value of f is attained at $x = -5$
- b) The minimum value of f is attained at $x = -2$
- c) The maximum value of f is $\frac{133}{2}$
- d) The minimum value of the derivative of f is attained at $x = -\frac{7}{2}$

1.1.35 Let $\mathbf{x}_1, \mathbf{x}_2, \mathbf{x}_3, \mathbf{x}_4, \mathbf{x}_5$ be a system of orthonormal vectors in $\mathbb{R}^{10}$. Consider the matrix $\mathbf{A} = \mathbf{x}_1\mathbf{x}_1^\top + \dots + \mathbf{x}_5\mathbf{x}_5^\top$. Which of the following statements is/are correct? (DA 2025)

- a) Singular values of $\mathbf{A}$ are also its eigenvalues
- b) Singular values of $\mathbf{A}$ are either 0 or 1
- c) Determinant of $\mathbf{A}$ is 1
- d) $\mathbf{A}$ is invertible

1.1.36 Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be a twice-differentiable function and suppose its second derivative satisfies $f''(x) > 0$ for all $x \in \mathbb{R}$. Which of the following statements is/are ALWAYS correct? (DA 2025)

- a) f has a local minima
- b) There does not exist x and $y, x \neq y$ such that $f'(x) = f'(y) = 0$
- c) f has at most one global minimum
- d) f has at most one local minimum

1.1.37 An $n \times n$ matrix $\mathbf{A}$ with real entries satisfies the property: $\|\mathbf{A}\mathbf{x}\|^2 = \|\mathbf{x}\|^2$, for all $\mathbf{x} \in \mathbb{R}^n$, where $\|\cdot\|$ denotes the Euclidean norm. Which of the following statements is/are ALWAYS correct? (DA 2025)

- a) $\mathbf{A}$ must be orthogonal
- b) $\mathbf{A} = \mathbf{I}$ where $\mathbf{I}$ denotes the identity matrix, is the only solution
- c) The eigenvalues of $\mathbf{A}$ are either +1 or -1
- d) $\mathbf{A}$ has full rank

1.1.38 Consider designing a linear binary classifier $f(\mathbf{x}) = \text{sign}(\mathbf{w}^\top \mathbf{x} + b)$, $\mathbf{x} \in \mathbb{R}^2$ on the following training data:

$$\text{Class-1} : \left\{ \begin{pmatrix} 2 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 2 \end{pmatrix}, \begin{pmatrix} 2 \\ 2 \end{pmatrix} \right\}, \quad \text{Class-2} : \left\{ \begin{pmatrix} 0 \\ 0 \end{pmatrix} \right\} \quad (1.1.38.1)$$

Hard-margin support vector machine (SVM) formulation is solved to obtain $\mathbf{w}$ and

b . Which of the following options is/are correct? (DA 2025)

- a) $\mathbf{w} = \begin{pmatrix} 4 \\ 4 \end{pmatrix}$ and $b = 1$
- b) The number of support vectors is 3
- c) The margin is $\sqrt{2}$
- d) Training accuracy is 98%

1.1.39 Consider a coin-toss experiment where the probability of head showing up is p . In

the i^{th} coin toss, let $X_i = 1$ if head appears, and $X_i = 0$ if tail appears. Consider

$$\hat{p} = \frac{1}{n} \sum_{i=1}^n X_i \quad (1.1.39.1)$$

where n is the total number of independent coin tosses. Which of the following statements is/are correct? (DA 2025)

- a) $E[\hat{p}] = p$
- b) $E[\hat{p}] = \frac{p}{n}$
- c) As n increases, variance of $\hat{p}$ decreases
- d) Variance of $\hat{p}$ does not depend on n

1.1.40 Let G be a simple, unweighted, and undirected graph. A subset of the vertices and edges of G are shown below.

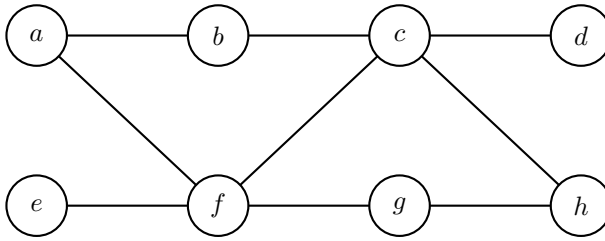


Fig. 1.1.40.1

It is given that $a - b - c - d$ is a shortest path between a and d ; $e - f - g - h$ is a shortest path between e and h ; $a - f - c - h$ is a shortest path between a and h . Which of the following is/are NOT the edges of G ? (DA 2025)

- a) (b, d)
- b) (b, g)
- c) (b, h)
- d) (e, g)

1.1.41 Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be such that $|f(x) - f(y)| \leq (x - y)^2$ for all $x, y \in \mathbb{R}$. Then

$$f(1) - f(0) = \underline{\hspace{2cm}} \quad (1.1.41.1)$$

(DA 2025)

1.1.42 Let $D = \{\mathbf{x}^{(1)}, \dots, \mathbf{x}^{(n)}\}$ be a dataset of n observations where each $\mathbf{x}^{(i)} \in \mathbb{R}^{100}$. It is given that $\sum_{i=1}^n \mathbf{x}^{(i)} = \mathbf{0}$. The covariance matrix computed from D has eigenvalues $\lambda_i = 100^{2-i}$, $1 \leq i \leq 100$. Let $\mathbf{u} \in \mathbb{R}^{100}$ be the direction of maximum variance with $\mathbf{u}^\top \mathbf{u} = 1$. The value of

$$\frac{1}{n} \sum_{i=1}^n \left(\mathbf{u}^\top \mathbf{x}^{(i)} \right)^2 = \underline{\hspace{2cm}} \quad (1.1.42.1)$$

(DA 2025)

1.1.43 A bag contains 5 white balls and 10 black balls. In a random experiment, n balls are drawn from the bag one at a time with replacement. Let S_n denote the total number of black balls drawn in the experiment. The expectation of S_{100} denoted by $E[S_{100}] = \underline{\hspace{2cm}}$. (DA 2025)

1.1.44 Consider the following tables, Loan and Borrower, of a bank.

Loan			Borrower	
loan_num	branch_name	amount	customer_name	loan_num
L11	Banjara Hills	90000	Anand	L11
L14	Kondapur	50000	Karteek	L11
L15	SR Nagar	40000	Karteek	L14
L22	SR Nagar	25000	Ankita	L15
L23	Balanagar	80000	Gopal	L19
L25	Kondapur	70000	Karteek	L22
L19	SR Nagar	65000	Karteek	L23
			Sunil	L23
			Sunil	L25

Fig. 1.1.44.1

Query: $\pi_{\text{branch_name}, \text{customer_name}}(\text{Loan} \bowtie \text{Borrower}) \div \pi_{\text{branch_name}}(\text{Loan})$ where $\bowtie$ denotes natural join. The number of tuples returned by the above relational algebra query is _____.

(DA 2025)

1.1.45 Consider the following Python code snippet.

```
def f(a, b):
    if (a == 0):
        return b
    if (a % 2 == 1):
        return 2 * f((a - 1) / 2, b)
    return b + f(a - 1, b)

print(f(15, 10))
```

The value printed by the code snippet is _____.

(DA 2025)

1.1.46 Consider the following pseudocode.

```
Create empty stack S
Set x = 0, flag = 0, sum = 0
Push x onto S

while (S is not empty) {
    if (flag equals 0){
        Set x = x + 1
        Push x onto S
    }
    if (x equals 8):
        Set flag = 1
        if (flag equals 1){
```

	<pre> x = Pop(S) if (x is odd): Pop(S) Set sum = sum + x } } Output sum </pre>
--	--

The value of sum output by a program executing the above pseudocode is _____.
(DA 2025)

- 1.1.47 Consider a directed graph $G = (V, E)$, where $V = \{0, 1, 2, \dots, 100\}$ and $E = \{(i, j) : 0 < j - i \leq 2 \text{ for all } i, j \in V\}$. Suppose the adjacency list of each vertex is in decreasing order of vertex number, and depth-first search (DFS) is performed at vertex 0. The number of vertices that will be discovered after vertex 50 is _____.
(DA 2025)